

Contractive Embeddings Between Weighted Bergman Spaces

A Full Literature-Implied Answer to arXiv:1701.06897

Status note for arXiv:1701.06897

12 August 2026

Status: full literature-implied answer. The answer is affirmative for every parameter in the question. The critical embedding is Kulikov's 2022 theorem; the extension to the full parameter wedge is an elementary radial stochastic-order argument. This note does not claim an original theorem.

1 The source question

For $s > 0$ and $\gamma > 1$, write

$$\|f\|_{A_\gamma^s}^s = \int_{\mathbb{D}} |f(z)|^s (\gamma - 1)(1 - |z|^2)^{\gamma-2} dA(z), \quad dA(z) = \frac{dx dy}{\pi},$$

and use the endpoint convention $A_1^s = H^s$. The source asks whether

$$\|f\|_{A_\beta^q} \leq \|f\|_{A_\alpha^p} \tag{1}$$

whenever

$$0 < p \leq q, \quad \alpha, \beta \geq 1, \quad \frac{q}{\beta} \leq \frac{p}{\alpha}.$$

The question begins on PDF p. 9 and continues on p. 10 [1].

2 Answer

Theorem. Inequality (1) holds for every analytic f and every set of parameters appearing in the source question.

3 Idea / proof intuition

Set

$$\beta_0 := \frac{\alpha q}{p}.$$

The source hypothesis is exactly $\beta \geq \beta_0$. Kulikov's theorem gives the hard step: contractivity at the critical line $p/\alpha = q/\beta_0$. It remains only to observe that, for a fixed exponent, increasing the

explains why this relationship is imposed in the statement of Theorem 1. The condition $\alpha \geq \alpha_0$ comes from [31, Thm. 1.3], but we do not know if it is necessary for the uniqueness of (14).

Question. For any $0 < p \leq q$ and $\alpha, \beta \geq 1$ such that $q/\beta \leq p/\alpha$, does the contractive inequality

$$\|f\|_{A_\beta^q(\mathbb{D})} \leq \|f\|_{A_\alpha^p(\mathbb{D})}$$

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hold? By Carleman's inequality and Theorem 1, this is true when $\beta = \alpha + n$ for some integer n , and either $\alpha = 1$ or $\alpha \geq \alpha_0$. We remark that it is easy to show, for example by computing with coefficients, that

$$\|f\|_{A_{2\alpha}^4(\mathbb{D})} \leq \|f\|_{A_\alpha^2(\mathbb{D})}$$

holds for every $\alpha \geq 1$.

2.3. Hypercontractivity of the Poisson kernel. For $r \in [0, 1]$, let P_r denote the operator defined on analytic functions in $\mathbb{D}$ by $P_r f(w) = f(rw)$. Clearly, if $r < 1$ it follows from (6) that P_r maps any $A_\alpha^p(\mathbb{D})$ into every $A_\beta^q(\mathbb{D})$. We are interested in knowing when this map is contractive.

Figure 1: The question in arXiv:1701.06897, PDF pp. 9–10.

Bergman weight parameter moves normalized radial mass toward the origin. Since the circular integral means of $|f|^q$ increase with the radius, this can only decrease the norm. Thus

$$A_\alpha^p \hookrightarrow A_{\beta_0}^q \hookrightarrow A_\beta^q$$

contractively.

4 Proof

For $s > 0$, define the circular integral mean

$$M_s(r; f) := \frac{1}{2\pi} \int_0^{2\pi} |f(re^{i\theta})|^s d\theta.$$

Because $|f|^s$ is subharmonic, $M_s(r; f)$ is nondecreasing in r .

We first record the promised monotonicity in the weight. If $\gamma > 1$ and X_γ is a random variable on $[0, 1]$ with density

$$(\gamma - 1)(1 - x)^{\gamma-2} dx,$$

then polar coordinates give

$$\|f\|_{A_\gamma^s}^s = \mathbb{E} M_s(\sqrt{X_\gamma}; f). \quad (2)$$

Moreover,

$$\mathbb{P}(X_\gamma > t) = (1 - t)^{\gamma-1} \quad (0 \leq t \leq 1).$$

Consequently, if $\gamma_2 \geq \gamma_1 > 1$, then X_{γ_2} is stochastically smaller than X_{γ_1} . Applying this to the increasing function $x \mapsto M_s(\sqrt{x}; f)$ in (2) yields

$$\|f\|_{A_{\gamma_2}^s} \leq \|f\|_{A_{\gamma_1}^s}. \quad (3)$$

If $\gamma_1 = 1$, the same conclusion follows from $M_s(r; f) \leq \|f\|_{H^s}^s$; hence (3) is valid with the Hardy endpoint as well.

Suppose first that $p < q$. If $\alpha > 1$, Kulikov's Corollary 1.3 applies to p, q, α, β_0 , since

$$\frac{p}{\alpha} = \frac{q}{\beta_0},$$

and gives

$$\|f\|_{A_{\beta_0}^q} \leq \|f\|_{A_\alpha^p}. \quad (4)$$

The same corollary contains the Hardy endpoint when $\alpha = 1$: indeed its rightmost inequality is $\|f\|_{A_{q/p}^q} \leq \|f\|_{H^p}$.

Theorem 1.2. *Let $G : [0, \infty) \rightarrow \mathbb{R}$ be a convex function. Then the maximum value of*

$$(1.4) \quad \int_{\mathbb{D}} G(|f(z)|^p (1 - |z|^2)^\alpha) dm(z)$$

is attained for $f(z) \equiv 1$, subject to the condition that $f \in A_\alpha^p$ and $\|f\|_{A_\alpha^p} = 1$.

Applying these theorems to the convex and increasing function $G(t) = t^s, s > 1$ we get that all the embeddings above between Hardy and Bergman spaces are contractions, that is we get the following corollary

Corollary 1.3. *For all $0 < p < q < \infty$ and $1 < \alpha < \beta < \infty$ with $\frac{p}{\alpha} = \frac{q}{\beta} = r$ for all functions f analytic in $\mathbb{D}$ we have*

$$\|f\|_{A_\beta^q} \leq \|f\|_{A_\alpha^p} \leq \|f\|_{H^r}$$

with equality for $f(z) \equiv c$ for $c \in \mathbb{C}$.

Figure 2: Kulikov's critical-line contraction and Hardy endpoint [2, Corollary 1.3].

Since the source hypothesis says $\beta \geq \beta_0$, combining (3) with (4) gives

$$\|f\|_{A_\beta^q} \leq \|f\|_{A_{\beta_0}^q} \leq \|f\|_{A_\alpha^p},$$

which is (1).

Finally, if $p = q$, the hypothesis is simply $\beta \geq \alpha$, so (1) follows directly from (3); the case $\alpha = \beta = 1$ is equality. This exhausts all endpoints and completes the proof. $\square$

5 Provenance and scope

Kulikov's paper appeared after the 2017 source and explicitly proves that the critical Hardy–Bergman and Bergman–Bergman embeddings are contractions. The passage from the critical line to all $\beta \geq \beta_0$ is the elementary monotonicity argument (3). The exact source question is therefore fully resolved, but the result is classified as *literature-implied*, not as an original result of this run.

This note answers only the displayed one-variable weighted Bergman embedding question. It makes no claim about the source's separate infinite-polydisk Carleson-measure question or its multiplicative-Hankel-form questions.

References

- [1] F. Bayart, O. F. Brevig, A. Haimi, J. Ortega-Cerdà, and K.-M. Perfekt, *Contractive inequalities for Bergman spaces and multiplicative Hankel forms*, arXiv:1701.06897v2, 2017.
- [2] A. Kulikov, *Functionals with extrema at reproducing kernels*, Geom. Funct. Anal. **32** (2022), 938–949; arXiv:2203.12349.